1. Introduction

The AI-Powered SOC Assistant enhances the Wazuh SIEM platform by providing automated analysis of security alerts. Instead of manually reviewing every event, the AI module enriches alerts with contextual information, helping analysts quickly understand potential threats and determine the appropriate response.

This guide explains how a Security Operations Center (SOC) analyst interacts with the system after deployment.

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2. Purpose

This guide enables users to:

- Access the Wazuh Dashboard
- Monitor endpoint security events
- Run the AI monitoring application
- Interpret AI-generated analysis
- Respond effectively to security incidents

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3. System Overview

The deployed solution consists of:

```
Windows Endpoint
      │
      ▼
Wazuh Agent
      │
      ▼
Wazuh Manager
      │
      ▼
alerts.json
      │
      ▼
AI Monitor
      │
      ▼
AI Analyzer
```

│
▼
SOC Analyst

The Wazuh Manager collects security events, while the AI module continuously monitors alerts and provides enriched security analysis.

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4. User Roles

Role	Responsibility
SOC Analyst	Monitor alerts, review AI analysis, investigate incidents
System Administrator	Manage Wazuh deployment, Docker services, and configurations
Security Administrator	Maintain detection rules and overall system security

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5. Accessing the Wazuh Dashboard

1. Open a web browser.
2. Navigate to:

https://<Manager-IP>

3. Log in using your administrator credentials.
4. Verify that the dashboard loads successfully.

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6. Viewing Security Alerts

1. From the Dashboard, open the **Security Events** or **Alerts** section.
2. Review recent alerts.
3. Click an alert to view detailed information, including:
    - Timestamp
    - Agent Name
    - Rule ID
    - Severity Level

- Description

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7. Understanding Alert Details

Each Wazuh alert provides:

Field	Description
Timestamp	Time the event occurred
Agent Name	Endpoint that generated the event
Rule ID	Detection rule triggered
Severity	Alert severity assigned by Wazuh
Description	Explanation of the detected event

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8. Running the AI Monitor

The AI Monitor continuously checks for new Wazuh alerts.

Step 1

Open a terminal on the Kali Linux host.

Step 2

Navigate to the project directory:

cd ~/AI_Wazuh_Lab

Step 3

Activate the virtual environment:

source venv/bin/activate

Step 4

Run the AI Monitor:

python ai_monitor.py

The monitor remains active and automatically processes new alerts as they are generated.

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9. Understanding AI Analysis

When a new alert is detected, the AI module enriches it with additional information.

Example output:

===== Latest Wazuh Alert =====

Agent: Balaganesh
Rule ID: 60106
Severity: 3
Description: Windows Logon Success

===== AI Security Analysis =====

Threat Category : Authentication

Risk Level : Low

Priority : P4

MITRE ATT&CK : T1078 - Valid Accounts

Executive Summary:
A successful Windows logon was detected. The activity appears to be a legitimate authentication event with no immediate indicators of compromise.

Recommendations:
• Verify the user account if unexpected.
• Continue monitoring for unusual login patterns.
• Correlate with additional events if necessary.

AI Confidence Score : 92%

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10. AI Output Fields

Field	Description
Threat Category	Classification of the security event
Risk Level	Low, Medium, High, or Critical

Field	Description
Priority	Incident priority (P1–P4)
MITRE ATT&CK	Relevant attack technique
Executive Summary	AI-generated explanation of the event
Recommendations	Suggested actions for the analyst
Confidence Score	Confidence level in the analysis

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11. Responding to Security Alerts

When reviewing AI analysis:

1. Confirm the affected endpoint.
2. Review the AI-generated executive summary.
3. Assess the risk level.
4. Follow the recommended remediation steps.
5. Investigate related events if required.
6. Escalate high-priority incidents according to organizational procedures.

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12. Best Practices

- Keep the AI Monitor running during active monitoring.
- Verify unexpected login events.
- Investigate repeated alerts from the same endpoint.
- Regularly review AI recommendations.
- Update Wazuh and Python dependencies when required.

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13. Troubleshooting

Issue	Resolution
Dashboard not loading	Verify Docker containers and Manager availability

Issue	Resolution
AI Monitor not detecting alerts	Confirm the correct alerts.json path and container name
Agent disconnected	Restart the Wazuh Agent and verify connectivity
JSON parsing error	Wait for the next complete alert entry
AI analysis unavailable	Check AI module configuration and dependencies

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14. Frequently Asked Questions (FAQ)

Q1. What is the purpose of the AI Monitor?

It continuously monitors Wazuh alerts and sends them to the AI Analyzer for enrichment.

Q2. Does the AI replace Wazuh?

No. Wazuh performs detection, while the AI provides additional context and recommendations.

Q3. What happens if the AI module is unavailable?

Wazuh continues to generate alerts. Analysts can still investigate events manually.

Q4. Can the AI monitor multiple endpoints?

Yes. As long as the endpoints report to the Wazuh Manager, the AI can analyze alerts generated from those systems.

Q5. Can additional AI models be integrated?

Yes. The modular design supports future integration with local LLMs, GPT-based services, or other AI models.

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15. Conclusion

The AI-Powered SOC Assistant simplifies the work of SOC analysts by automatically enriching Wazuh alerts with actionable insights. It helps reduce manual analysis time, improves understanding of security events, and supports faster, more informed incident response while maintaining compatibility with the existing Wazuh SIEM infrastructure.